Table 10. Bounded runtime and memory benchmark

Panel A. Timing

Workload group	Method	Phase	Status	Repetitions	Warmup repetitions	Median time (ms)	Minimum time (ms)	Maximum time (ms)
matched coherent statevector	NumPy dense six-state propagation	execution	measured	7	2	0.0005	0.0005	0.0007
matched coherent statevector	Qiskit Statevector dense unitary	circuit construction	measured	7	2	0.0331	0.0310	0.0364
matched coherent statevector	Qiskit Statevector dense unitary	execution	measured	7	2	0.0405	0.0356	0.0416
matched coherent statevector	Qiskit Statevector dense unitary	post-processing	measured	7	2	0.0034	0.0032	0.0040
matched coherent statevector	Qiskit Statevector dense unitary	transpilation	not applicable	0	0	Not available	Not available	Not available
matched coherent statevector	Qiskit Statevector dense unitary	sampling	not applicable	0	0	Not available	Not available	Not available
matched coherent statevector	quantum hardware	hardware execution	not applicable	0	0	Not available	Not available	Not available
matched classical open system	constant-generator matrix exponential	execution	measured	7	2	1.6518	1.6405	1.6720
matched classical open system	adaptive Dormand-Prince RK5(4)	execution	measured	7	2	68.0221	67.6639	68.9592

Timings are environment-specific medians for the retained six-state model, not a scaling or speed-advantage claim. Process peak RSS is a whole-process high-water mark; theoretical array bytes are lower-bound storage estimates. Presentation values are rounded. All 9 source rows and all 27 source columns remain in table10_runtime_memory_benchmark.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 10. Bounded runtime and memory benchmark

Panel B. Memory measurement and estimate

Workload group	Method	Matched problem	Timing boundary	Process peak RSS (MiB)	Measured peak rss limitation	Core-array estimate (bytes)	Theoretical storage scope
matched coherent s tatevector	NumPy dense six-state propagation	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	precomputed 6x6 unitary multiplied by supplied 6-element statevector	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	768	one 6x6 complex unitary plus input and output statevectors
matched coherent s tatevector	Qiskit Statevector dense unitary	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	construct QuantumCircuit and append precomputed 8x8 UnitaryGate	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	1024	embedded 8x8 complex unitary only; excludes Qiskit object overhead
matched coherent s tatevector	Qiskit Statevector dense unitary	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	Statevector(input).evo lve(preconstructed circuit); no transpilation or shots	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	1280	embedded unitary plus input and output 8-element statevectors
matched coherent s tatevector	Qiskit Statevector dense unitary	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	compute D/Q expectations and unused-state leakage from exact statevector	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	2176	output statevector plus embedded D/Q projector arrays
matched coherent s tatevector	Qiskit Statevector dense unitary	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	not applicable: Statevector evolves the inserted dense unitary directly	Not available	whole Python-process high-water mark; not incremental or attributable to one operation	0	not applicable
matched coherent s tatevector	Qiskit Statevector dense unitary	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	not applicable: no finite-shot measurement is executed	Not available	whole Python-process high-water mark; not incremental or attributable to one operation	0	not applicable

Timings are environment-specific medians for the retained six-state model, not a scaling or speed-advantage claim. Process peak RSS is a whole-process high-water mark; theoretical array bytes are lower-bound storage estimates. Presentation values are rounded. All 9 source rows and all 27 source columns remain in table10_runtime_memory_benchmark.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 10. Bounded runtime and memory benchmark

Panel B. Memory measurement and estimate (continued)

Workload group	Method	Matched problem	Timing boundary	Process peak RSS (MiB)	Measured peak rss limitation	Core-array estimate (bytes)	Theoretical storage scope
matched coherent s tatevector	quantum hardware	same supplied normalized six-state vector, dense U=exp(-iHt), t=1/kref=1e-06 s	not applicable: no quantum hardware is used	Not available	whole Python-process high-water mark; not incremental or attributable to one operation	0	not applicable
matched classical open system	constant-generator matrix exponential	same 39-component open-system IVP; mixture pD=0.75; matched H, reaction, relaxation, escape, 31 output times, t=4/kref	complete solver call including generator construction, propagation, and outputs	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	43680	39x39 generator plus 39-component stored trajectory; lower bound
matched classical open system	adaptive Dormand-Prince RK5(4)	same 39-component open-system IVP; mixture pD=0.75; matched H, reaction, relaxation, escape, 31 output times, t=4/kref	complete solver call including adaptive integration and output assembly	389.2969	whole Python-process high-water mark; not incremental or attributable to one operation	24336	stored 39-component trajectory plus eight stage/state vectors; lower bound

Timings are environment-specific medians for the retained six-state model, not a scaling or speed-advantage claim. Process peak RSS is a whole-process high-water mark; theoretical array bytes are lower-bound storage estimates. Presentation values are rounded. All 9 source rows and all 27 source columns remain in table10_runtime_memory_benchmark.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 10. Bounded runtime and memory benchmark

Panel C. Execution context

Workload group	Method	Hardware	Software	Transpilation status	Hardware execution status	Sampling status	Interpretation
matched coherent statevector	NumPy dense six-state propagation	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched coherent statevector	Qiskit Statevector dense unitary	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched coherent statevector	Qiskit Statevector dense unitary	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched coherent statevector	Qiskit Statevector dense unitary	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched coherent statevector	Qiskit Statevector dense unitary	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched coherent statevector	Qiskit Statevector dense unitary	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference

Timings are environment-specific medians for the retained six-state model, not a scaling or speed-advantage claim. Process peak RSS is a whole-process high-water mark; theoretical array bytes are lower-bound storage estimates. Presentation values are rounded. All 9 source rows and all 27 source columns remain in table10_runtime_memory_benchmark.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 10. Bounded runtime and memory benchmark

Panel C. Execution context (continued)

Workload group	Method	Hardware	Software	Transpilation status	Hardware execution status	Sampling status	Interpretation
matched coherent statevector	quantum hardware	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable unless stated	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched classical open system	constant-generator matrix exponential	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference
matched classical open system	adaptive Dormand-Prince RK5(4)	Darwin arm64; logical CPUs=10; macOS-15.7.3-arm64-arm-64bit	Python 3.12.2; NumPy 2.3.2; Qiskit 2.1.1	not applicable	not used	not used; exact statevectors and expectations	bounded reproducibility benchmark only; no speed, efficiency, scaling, or quantum-advantage inference

Timings are environment-specific medians for the retained six-state model, not a scaling or speed-advantage claim. Process peak RSS is a whole-process high-water mark; theoretical array bytes are lower-bound storage estimates. Presentation values are rounded. All 9 source rows and all 27 source columns remain in table10_runtime_memory_benchmark.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.